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AI-Powered Crop Yield and Health Monitoring System

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Abstract*— Monitoring plant health is essential to sustainable agriculture since it allows for prompt action to increase crop yield and reduce losses. This research uses cutting-edge ML & DL approaches to identify, categorise, and forecast plant health in an AI-driven manner. Crop yield prediction, crop health identification/classification, and crop health prediction are the three interconnected modules that make up the framework. To precisely detect and categorise plant health issues, a logistic regression classifier is used in conjunction with a pretrained ResNet-50 model for feature extraction. To guarantee dependability in a variety of settings, the system is trained on a carefully selected dataset using strong preprocessing and data augmentation techniques. Evaluation criteria that show how well the model identifies plant health states. Furthermore, visualisation techniques like feature map analysis offer interpretability into the model's decision-making process. The initiative promotes precision agriculture and sustainable farming methods by providing a scalable solution for real-time plant health monitoring. In the future, the system will be expanded to include a wider range of crops and weather circumstances, and models for edge deployment will be improved.

Keywords— **Plant health monitoring, crop yield prediction, crop health classification, ResNet-50, logistic regression, sustainable agriculture, precision farming, machine learning, deep learning, plant disease detection, feature extraction, data augmentation, t-SNE visualization, real-time monitoring, AI in agriculture.**

I. INTRODUCTION

The development of intelligent systems that can solve important agricultural issues has been made possible by the quick advances in AI. Among these, monitoring plant health is essential for guaranteeing maximum agricultural yield and reducing losses from illnesses and environmental stressors. Manual observations, which are labour-intensive, time-consuming, and prone to errors, are frequently used in traditional crop health monitoring techniques. AI-driven solutions, on the other hand, offer accurate, scalable, and real-time plant health analysis, giving farmers useful information.

This project, AI-Driven Plant Health Monitoring System, aims to integrate advanced AI techniques into a unified framework for comprehensive crop management. The system is designed with three core modules:

Crop Yield Prediction: This module forecasts possible crop yields by utilising historical data and environmental factors, assisting farmers in making well-informed decisions about the management and distribution of resources. Identification and Classification of Crop Health: This lesson uses DL & CV techniques to identify and categorise plant health situations, including healthy plants and those impacted by pests, diseases, or nutritional deficits. Crop Health Forecast: Based on present circumstances and trends, this module uses predictive modelling and time-series analysis to anticipate the health status of crops in the future.

The project provides a thorough solution that covers several facets of plant health management by combining these components. To guarantee solid and trustworthy analysis, the system makes use of data from a variety of sources, such as sensor data, satellite photography, and human inputs. Adopting an AI-driven strategy improves agricultural techniques' accuracy and efficiency while also promoting sustainable farming by cutting waste, making the most of inputs, and strengthening resistance to environmental shocks.



Fig 1: Crop Health Monitoring [21]

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II. LITERATURE REVIEW

The combination of AI & ML has led to major breakthroughs in the fields of plant health monitoring and agricultural production prediction. Predictive modelling and real-time analysis have been used in several studies to improve agricultural output using a variety of approaches and instruments. Key contributions in this field are collected in this literature review.

A thorough evaluation of the literature on ML methods for agricultural production prediction was carried out by Van Klompenburg et al. [1]. In order to create precise and trustworthy prediction models, they emphasised the significance of feature selection, data preprocessing, and model assessment. Their research included a thorough analysis of the most recent techniques and difficulties in predicting crop productivity. Similarly, with an emphasis on palm oil output, Rashid et al. [2] provided a thorough evaluation of ML-based crop yield prediction. They underlined how important it is to include environmental and climatic elements in prediction models. In addition to outlining the advantages and disadvantages of different machine learning algorithms, their study suggested future lines of inquiry to increase prediction accuracy.

By using machine learning approaches to examine soil, weather, and crop-specific data, Gopal and Bhargavi [4] presented a revolutionary method for effectively predicting crop production. Their research offered insights into practical uses of predictive models and emphasised the significance of merging data from many sources to provide precise forecasts. A thorough review of agricultural yield prediction utilising remote sensing and deep learning was carried out by Muruganantham et al. [5]. Their research demonstrated the expanding use of remote sensing and satellite imaging in agricultural applications. They underlined that in order to increase the prediction models' resilience, geographical and temporal data must be integrated.

A machine learning approach for identifying crop health in smart agricultural systems was presented by Shukla et al. [6]. They classified plant health states and identified disease signs by analysing sensor data. This study demonstrated ML's potential for early disease diagnosis, which might lower crop losses and enhance management techniques. Pham et al. [7] looked at the use of vegetation health indices based on satellites to improve agricultural output forecast. Their research showed how satellite data may be used by machine learning systems to accurately forecast crop yields. The advantages of real-time monitoring and the scalability of these methods for expansive agricultural landscapes were emphasised by the writers.

Petersen [8] investigated the use of MODIS relative vegetation health data throughout Africa for real-time agricultural yield prediction. The study shed light on how remote sensing technology may be used to track and forecast agricultural performance on a continental level. Lastly, an integrated system for automated disease prediction and plant growth monitoring was created by Mishra and Deepa [10]. Their method offered a comprehensive approach to agricultural management by fusing disease detection algorithms with crop forecast models. The study underlined how crucial automation and integration are to solving today's agriculture problems.

When taken as a whole, this research show how ML & DL may revolutionise the agricultural industry. Researchers have cleared the path for precise crop output forecasts, effective plant health monitoring, and sustainable agricultural methods by utilising a variety of data sources and sophisticated algorithms. There is a solid basis for future innovation in this area thanks to the potential and difficulties this body of work highlights.

III. METHODOLOGY

To develop a comprehensive system for plant health monitoring, the research focused on integrating advanced data processing techniques with machine learning models. The system was structured into three interconnected modules: crop yield prediction, crop health identification/classification, and crop health prediction. Below is an account of how each component was designed and implemented:

3.1 Crop Yield Prediction Module:

3.1.1 Data Preprocessing:

The dataset is loaded and cleaned by handling missing values using the dropna() function. Categorical variables such as State_Name, District_Name, Season, and Crop are encoded using Label Encoding.

3.1.2 Exploratory Data Analysis:

Data visualizations were created to analyze crop production distribution by state, crop, season, and area using bar plots, box plots, and heatmaps.Feature Selection: The dataset was divided into independent variables and the dependent variable (Production).

3.1.3 Model Training:

The dataset was split into training and testing sets (75% training, 25% testing). Two regression models, RFR and DTR, were used for training. Hyperparameters of the models were optimized for performance.

3.1.4 Evaluation Metrics:

The models were evaluated using metrics such as R^2 score and accuracy percentage. Random Forest Regressor achieved a high performance with a strong R^2 score.

3.1.5 Visualization:

Predictions were compared against actual values using scatter plots to analyze the model's predictive capabilities.

3.1.6 Model Deployment:

The Random Forest model was serialized into a .pkl file using Python's pickle module for future deployment in production environments.

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The process flow of the Crop Yield Prediction Module is illustrated in Figure 2.

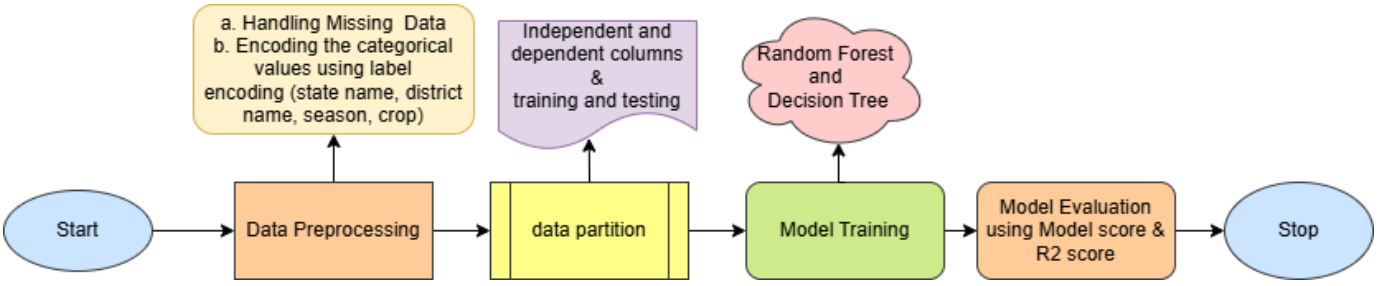


Figure 2: Crop Yield Prediction Module

3.2 Crop Health Prediction Module:

3.2.1 Dataset Preparation:

A plant health dataset containing temporal and environmental features was loaded and explored. Data cleaning included handling missing values, duplicate rows, and statistical outlier detection using Z-score. Categorical variables, including Plant_Health_Status, were encoded into numerical values using a custom mapping.

3.2.2 Feature Engineering:

Temporal features were analyzed by resampling data into daily and weekly aggregates for key environmental metrics (e.g., soil moisture, temperature). Correlation analysis identified features significantly related to plant health.

3.2.3 Data Splitting and Scaling:

The dataset was split into training and testing subsets (80:20 split), ensuring class balance using stratified sampling. Features were standardized using StandardScaler for improved model performance.

3.2.4 Model Training and Evaluation:

Six classifiers—KNN, DT, RF, GB, LR, and SVM—were trained on the data. Models were evaluated using metrics. The best-performing model was identified based on AUC and accuracy scores.

3.2.5 Visualization:

Data distribution and feature importance were visualized using histograms, heatmaps, and bar charts. Model predictions were compared against actual values with distribution plots and confusion matrices.

3.2.6 Result Summary:

A comprehensive evaluation summary of all models was generated, highlighting the best model. Feature importance and prediction distributions were plotted to interpret the model's decisions.

The process flow of the Crop Health Prediction Module is illustrated in Figure 3.

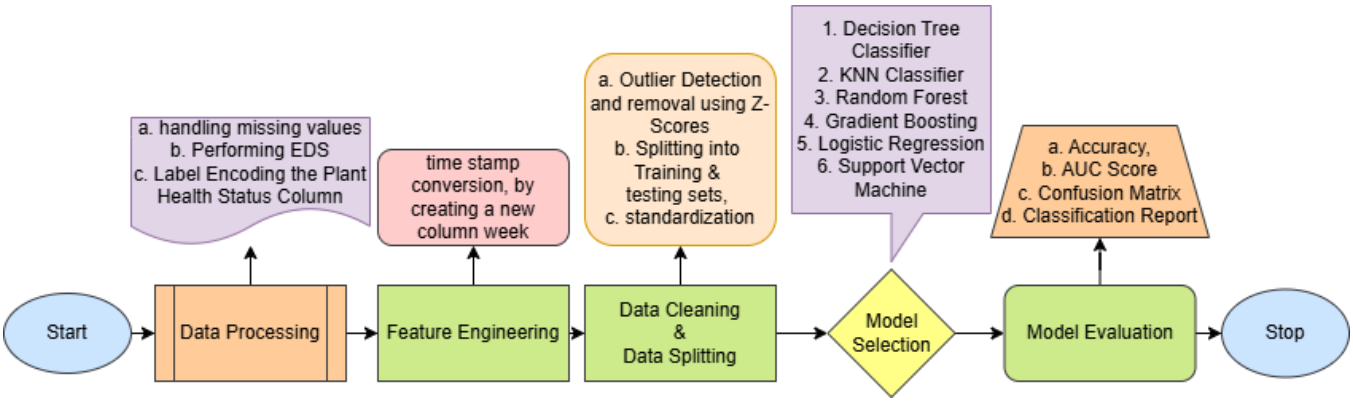


Figure 3: Crop Health Prediction

3.3 Crop Health Identification/Classification Module:

3.3.1 Data Preprocessing:

Images were resized to 224×224 pixels, transformed into tensors and normalised using the ImageNet dataset's mean and standard deviation values.

3.3.2 Dataset Preparation:

Train and test datasets were created using the ImageFolder class. Data loaders were implemented with a batch size of 32 for efficient loading.

3.3.3 Model Selection:

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A pre-trained ResNet-50 model was utilized for feature extraction.

3.3.4 Feature Extraction:

Features were extracted from the image datasets using ResNet-50, and flattened to create input vectors for the classifier.

3.3.5 Classifier Training:

A logistic regression model was implemented with a linear layer for multi-class classification. The model was trained for 50 epochs using cross-entropy loss and the SGD optimizer.

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3.3.6 Model Evaluation:

Performance metrics such as accuracy, and confusion matrix were calculated using the test dataset.

3.3.7 Visualization:

Class distribution, feature maps, and embeddings were visualized using t-SNE and feature extraction methods to understand model performance.

3.3.8 Model Saving:

The trained logistic regression model was saved for future inference.

The process flow of the Crop Health Identification Module is illustrated in Figure 4.

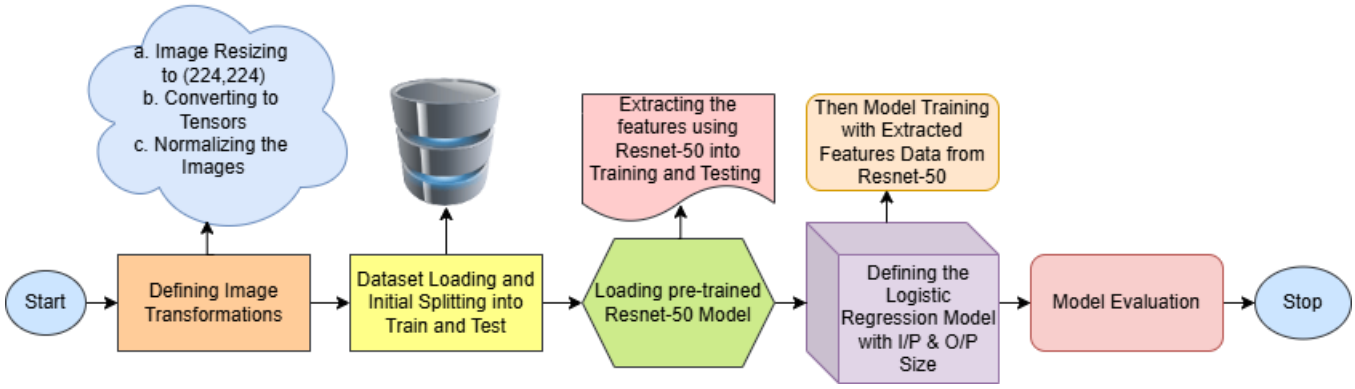


Figure 4: Crop Health Identification/Classification

3.4 Dataset Description

3.4.1 Dataset Overview

The project leverages three distinct datasets to address the various components of plant health monitoring and crop yield prediction, including crop yield prediction, crop health prediction, and crop health classification. Below is a detailed analysis and description of each dataset used in this research.

Note: The dataset used for this study was obtained from Kaggle, which is already well-processed and free from missing values or inconsistencies.

3.4.2 Crop Yield Prediction Dataset

For the crop yield prediction module, the dataset used contains information about various crops across different states and districts in India. The dataset includes historical data on crop production and area harvested, providing a solid foundation for yield prediction models. The structure of the dataset consists of the following columns:

a. State_Name: The name of the state in which the crop is grown (e.g., Andaman and Nicobar Islands).
b. District_Name: The specific district within the state.
c. Season: The crop-growing season (e.g., Kharif, Rabi).
d. Crop: The type of crop grown (e.g., Arecanut).
e. Area: The total area (in hectares) used for cultivation of the crop.
f. Production: The total yield (in tons) of the crop.

This dataset offers a wealth of information for building predictive models that can estimate crop yields based on historical data. It provides insights into crop behavior, seasonal variations, and geographical factors, which are essential for forecasting future production. The dataset is instrumental for developing machine learning models aimed at improving agricultural planning and resource management.

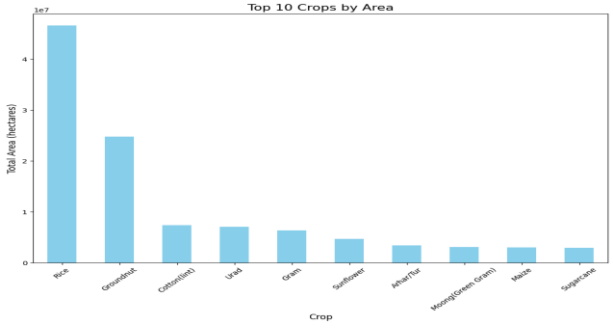


Figure 5: Top 10 Crops by Area

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3.4.3 Crop Health Prediction Dataset

The dataset for crop health prediction contains environmental and soil parameters that are crucial for monitoring plant health. It includes sensor data and plant health status information, which are essential for determining the overall well-being of crops. The dataset consists of the following columns:

Timestamp, Plant name, Soil Moisture, Ambient Temperature, Soil Temperature Humidity, Light Intensity, Soil pH, Nitrogen Level, Phosphorus Level, Potassium Level, Chlorophyll Content, Electrochemical Signal, Plant Health Status.

Figure 6: Plant Health Prediction Dataset Columns

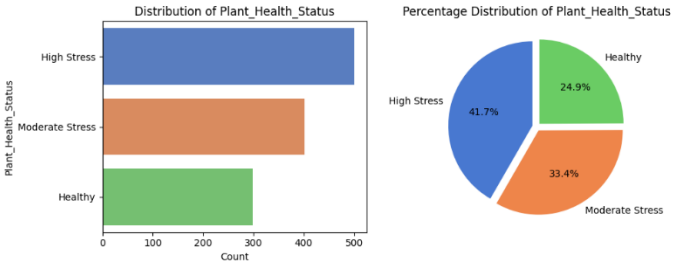


Figure 7: Plant Health Status Visualization

3.4.4 Crop Health Identification/Classification Dataset

Seventy thousand high-quality photos of both healthy and ill plant leaves make up the crop health identification and classification collection. These photos are divided into classes that depict distinct plant illnesses and situations that are beneficial for different kinds of plants. This dataset is unique in its diversity, covering nine different species of plants, each with its own set of disease classes. The classes for Apple species are as follows:

a. Apple Dataset: Apple Scab, Black Rot, Cedar Apple Rust, Healthy
b. Similarly, for Tomato, Bell Pepper, Strawberry, Corn (Maize), Peach, Grape, Cherry, Potato.

3.5 Understanding Algorithms Usage:

The selection of algorithms for this project was guided by the specific requirements of each module:

a. **Crop Yield Prediction**: Random Forest and Decision Tree algorithms were chosen for their capability to manage structured datasets and capture intricate patterns in agricultural data.
b. **Crop Health Prediction**: Multiple machines learning models, including KNN, DT, RF, GB, LR, and SVM, were implemented to analyse plant health using environmental and sensor data, ensuring a robust evaluation of different methodologies.
c. **Crop Health Identification/Classification**: ResNet-50, a deep learning model, and logistic regression was leveraged for its strength in handling high-dimensional image datasets and accurately classifying plant health across various species and conditions.

The diverse algorithmic approaches align with the distinct characteristics of each problem domain, ensuring an optimal solution framework for agricultural analytics.

3.6 Novelty of AI-Powered Crop Yield and Health Monitoring System

The proposed AI-powered Crop Yield and Health Monitoring System introduces several novel contributions that distinguish it from existing research in agricultural technology and precision farming. These contributions include the integration of advanced machine learning (ML) and deep learning (DL) techniques, an innovative modular framework, robust data preprocessing, and real-time interpretability through visualization. The key novel aspects of this research are outlined below:

a. **Comprehensive Multi-Module Framework:**
Unlike previous studies that focus on isolated aspects of crop health monitoring, this research presents an integrated system combining crop yield prediction, crop health classification, and crop health forecasting. The seamless interaction between these three modules enhances predictive capabilities, allowing for more informed agricultural decision-making.

b. **Hybrid Model Approach for Crop Health Identification:**
The combination of a logistic regression classifier with a pre-trained ResNet-50 model for feature extraction is a unique approach to plant health classification. This hybrid model ensures efficient feature learning and classification, improving accuracy in detecting and categorizing plant health conditions.

c. **Robust Model Evaluation and Comparative Analysis:**
Multiple ML models (KNN, Decision Tree, Random Forest, Gradient Boosting, Logistic Regression, and SVM) are employed for crop health prediction, ensuring robust benchmarking and selection of the best-performing model. The research offers a comparative performance analysis based on accuracy, AUC scores, and confusion matrix evaluations, providing insights into model effectiveness.

d. **Scalability and Real-Time Deployment Considerations:**
The modular design of the system allows for future scalability, making it adaptable to various crops and environmental conditions. The trained models are serialized and optimized for real-time deployment, ensuring practical applicability in precision farming.

e. **Integration of Multiple Data Sources for Enhanced Accuracy:**
The study utilizes a combination of structured crop yield datasets, environmental sensor data, and high-resolution plant leaf images to improve prediction accuracy. This multi-source data integration strategy strengthens the reliability of the proposed system compared to conventional single-data-source approaches.

By incorporating these novel aspects, the AI-powered Crop Yield and Health Monitoring System advances the field of precision agriculture, providing a scalable and intelligent

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solution for sustainable farming. The proposed framework not only enhances crop health monitoring but also contributes to efficient resource allocation and reduced agricultural losses due to undetected plant health issues.

IV. ARCHITECTURURAL DESCRIPTION

The project is designed with a modular architecture to address crop yield prediction, crop health prediction, and crop health identification/classification. Each module functions independently yet integrates seamlessly to provide comprehensive agricultural insights:

1. **Data Collection Layer**: This layer includes datasets for structured agricultural data, sensor data, and plant leaf images, forming the foundation for analysis.
2. **Data Preprocessing and Feature Engineering**: Data is cleaned, normalized, and transformed to extract meaningful features. For images, preprocessing involves resizing and augmenting to enhance model performance.
3. **Modelling Layer**:
   a. **Crop Yield Prediction**: Implements Decision Tree and Random Forest models to analyze structured crop production data.
   b. **Crop Health Prediction**: Employs multiple machine learning algorithms to predict plant health based on environmental and sensor data.
   c. **Crop Health Identification/Classification**: Utilizes ResNet-50 for precise classification of plant diseases and healthy conditions from image datasets.
4. **Integration and Decision Support**: The outputs from all modules are aggregated into a unified system to offer actionable insights for optimizing crop productivity and health management.
5. **Visualization and Deployment**: Results are presented through user-friendly visualizations, enabling farmers and stakeholders to make informed decisions.

This architecture ensures scalability, modularity, and effectiveness in addressing diverse agricultural challenges.

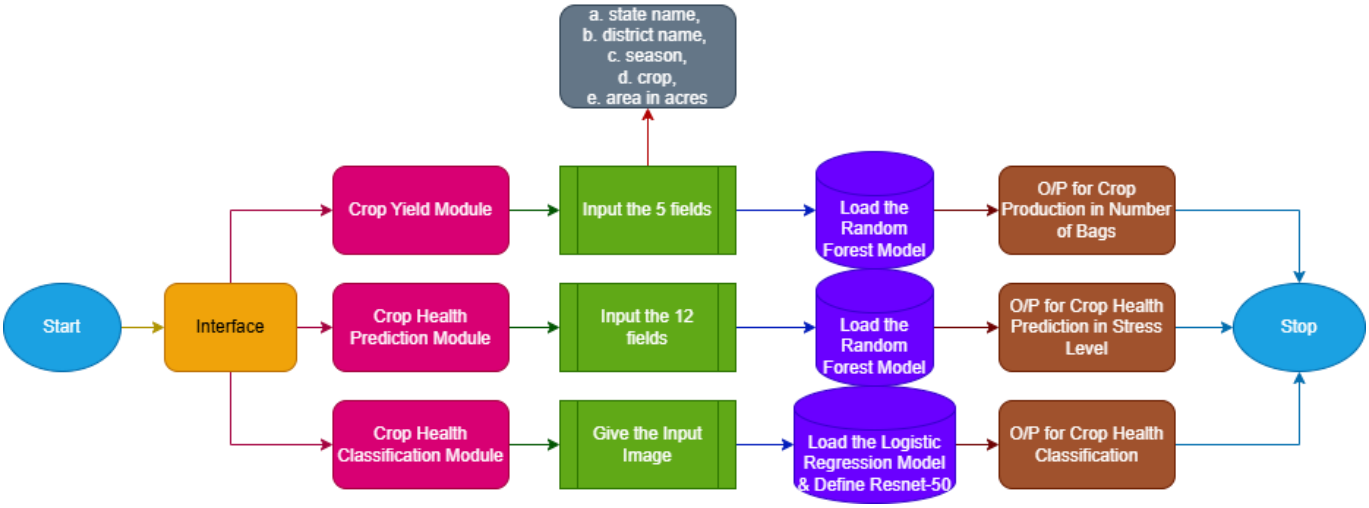


Figure 8: Overall Architecture

V. EXPERIMENTATION AND RESULTS

5.1 Crop Yield Prediction Results

Table **1** presents the **model performance** for crop yield prediction. The **Random Forest Regressor (RF)** outperformed the **Decision Tree Regressor (DT)**, achieving a **higher R^2 score**, which indicates better variance explanation in the crop yield data.

Table 1: Model Performance for Crop Yield

Model	Model Score	R2 Score
Random Forest	97.36%	97.36%
Decision Tree	95.05%	95.05%

Figure **9** and Figure **10** illustrate the **scatter plots** comparing the predicted and actual crop yield values for the **Random Forest** and **Decision Tree** models, respectively.

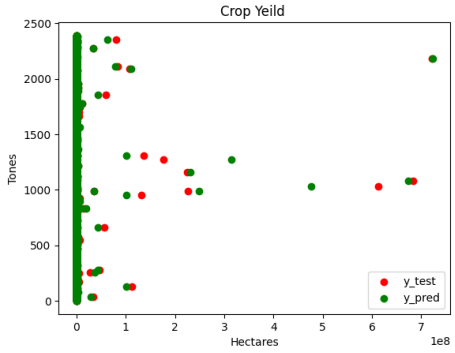


Fig 9: Scatter Plot: Random Forest Regressor

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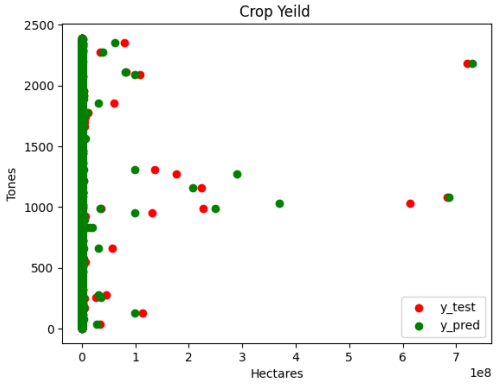


Fig 10: Scatter Plot: Decision Tree Regressor

These results indicate that **both models** achieved **high accuracy**, but the **Random Forest model demonstrated superior predictive performance** with a **higher R^2 score**, making it a more reliable choice for crop yield prediction.

5.2 Crop Health Prediction

For crop health prediction, the following models were evaluated based on accuracy and AUC (Area Under the Curve). **Table 2** presents the performance metrics of these models.

Table 2: Model Performance for Crop Health Prediction

S.No	Model	Accuracy	AUC
1	Decision Tree	0.991667	0.993472
2	Random Forest	0.987500	0.999664
3	Gradient Boosting	0.987500	0.993996
5	Support Vector Machine	0.808333	NaN
4	Logistic Regression	0.758333	0.902026
6	KNN Classifier	0.637500	0.813661

Based on **AUC**, the **Random Forest** model is the best-performing model with an AUC of 1.00, indicating excellent performance in distinguishing between the health statuses of plants.

The **Feature Importance** heatmap for the **Random Forest** model (Figure 11) illustrates the relative importance of various features in predicting crop health. Features that have high importance include **soil moisture, temperature, humidity, and light intensity**, indicating their significant role in determining plant health.

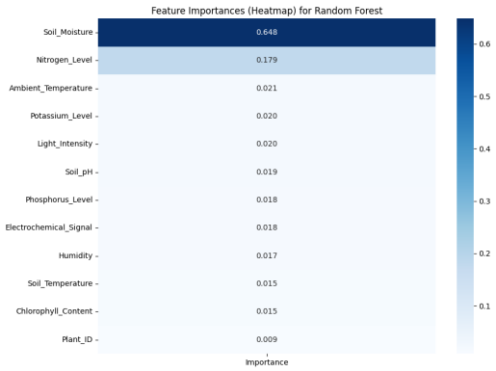


Figure 11: Feature Importance Heatmap: Random Forest

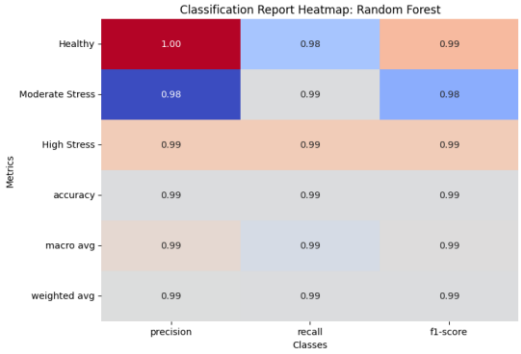


Figure 12: Classification Report Heatmap: Random Forest

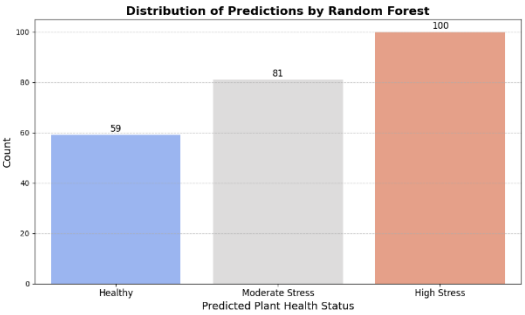


Figure 13: Distribution of Predictions by Random Forest

5.3 Crop Health Identification/Classification (Using ResNet-50 and Logistic Regression)

The model was trained for 50 epochs and the following results were achieved:

a. **Epoch [50] Loss:** 0.256
b. **Accuracy:** 92.86%
c. **Precision:** 92.92%
d. **Recall:** 92.86%
e. **F1 Score:** 92.80%

5.3.1 Confusion Matrix Analysis

The **confusion matrix** for the **Apple Dataset** in the crop health identification task is presented in **Figure 15**.

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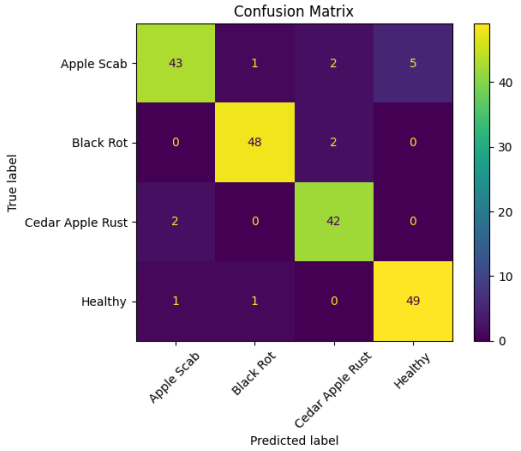


Figure 14: Confusion Matrix– Crop Health Identification Model

The results indicate a **high-performing classification model**, where the **ResNet-50 architecture combined with Logistic Regression** demonstrates **strong precision, recall, and F1 scores**, making it effective in detecting and classifying plant diseases.

Furthermore, **Figure 16** illustrates the classification of **plant leaves into different disease categories**, showcasing how the model differentiates between **various plant health conditions**.

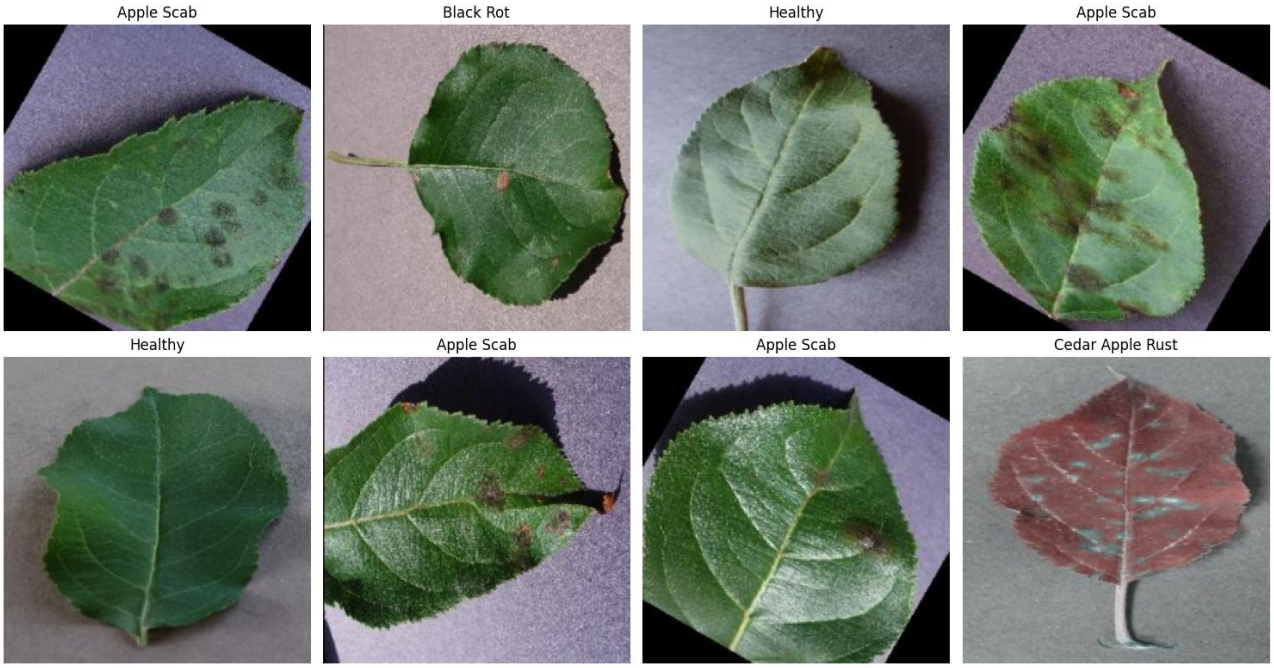


Figure 15: Classification of Plant Leaves into Different Classes

VI. CONCLUSION

This research presents a comprehensive approach to enhancing agricultural productivity and sustainability by leveraging advanced ML & DL. The integration of crop yield prediction, crop health prediction, and plant disease classification demonstrates the potential of technology in addressing critical challenges faced by the agricultural sector. By utilizing diverse datasets and implementing robust models, the project delivers accurate and actionable insights that can empower farmers to make informed decisions. The modular design ensures scalability and adaptability, making it suitable for various agricultural contexts. Overall, this research underscores the transformative role of data-driven solutions in modern farming, paving the way for smarter, more efficient agricultural practices.

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